

Patent-Ready Process Description: LEA-Cu-Mg-Al Hyper-Alloy

Title:

Method for Producing a Conductive Aluminum-Copper-Magnesium Alloy with Tunable Mechanical and Electrical Properties

Abstract (optional):

The invention relates to a method for producing a conductive aluminum-based alloy comprising copper and magnesium, designed to achieve a balance between electrical conductivity and mechanical strength. The process includes controlled melting, casting, thermal homogenization, deformation, solution treatment, quenching, and artificial aging, with defined temperature and time windows to tailor alloy performance.

Process Claims (formal language):

Claim 1:

A method for producing a conductive aluminum-copper-magnesium alloy, comprising the steps of:

- Melting a composition consisting of 93.5–96.5 wt% aluminum, 1.5–3.5 wt% copper, 1.0–2.0 wt% magnesium, and ≤ 0.2 wt% trace elements, at a temperature between 650 °C and 700 °C;
- Casting the molten alloy into billets or ingots at a temperature between 680 °C and 700 °C;
- Optionally performing homogenization annealing at 480 °C to 520 °C for a duration of 4 to 12 hours;
- Hot forming the cast alloy by extrusion or rolling at a temperature between 400 °C and 450 °C;
- Subjecting the formed alloy to solution heat treatment at 500 °C to 520 °C for 30 to 90 minutes;
- Quenching the alloy in water or air to retain a supersaturated solid solution;
- Performing artificial aging at 150 °C to 200 °C for 4 to 12 hours to adjust the balance between electrical conductivity and mechanical strength.

Technical Description (for specification section):

The process enables the production of a hyper-conductive aluminum alloy with tunable properties by adjusting copper and magnesium content and thermal

treatment parameters. The artificial aging step may be optimized for either:

- High-conductivity grade (LEA-90): longer aging time and higher temperature to reduce precipitate density and maximize electron mobility;
- Balanced grade (LEA-100): intermediate aging conditions for dual performance;

- High-strength grade (LEA-120): shorter aging time and lower temperature to retain fine precipitates and increase yield strength.

The alloy is suitable for applications in lightweight power transmission, thermal management, and structural electronics.

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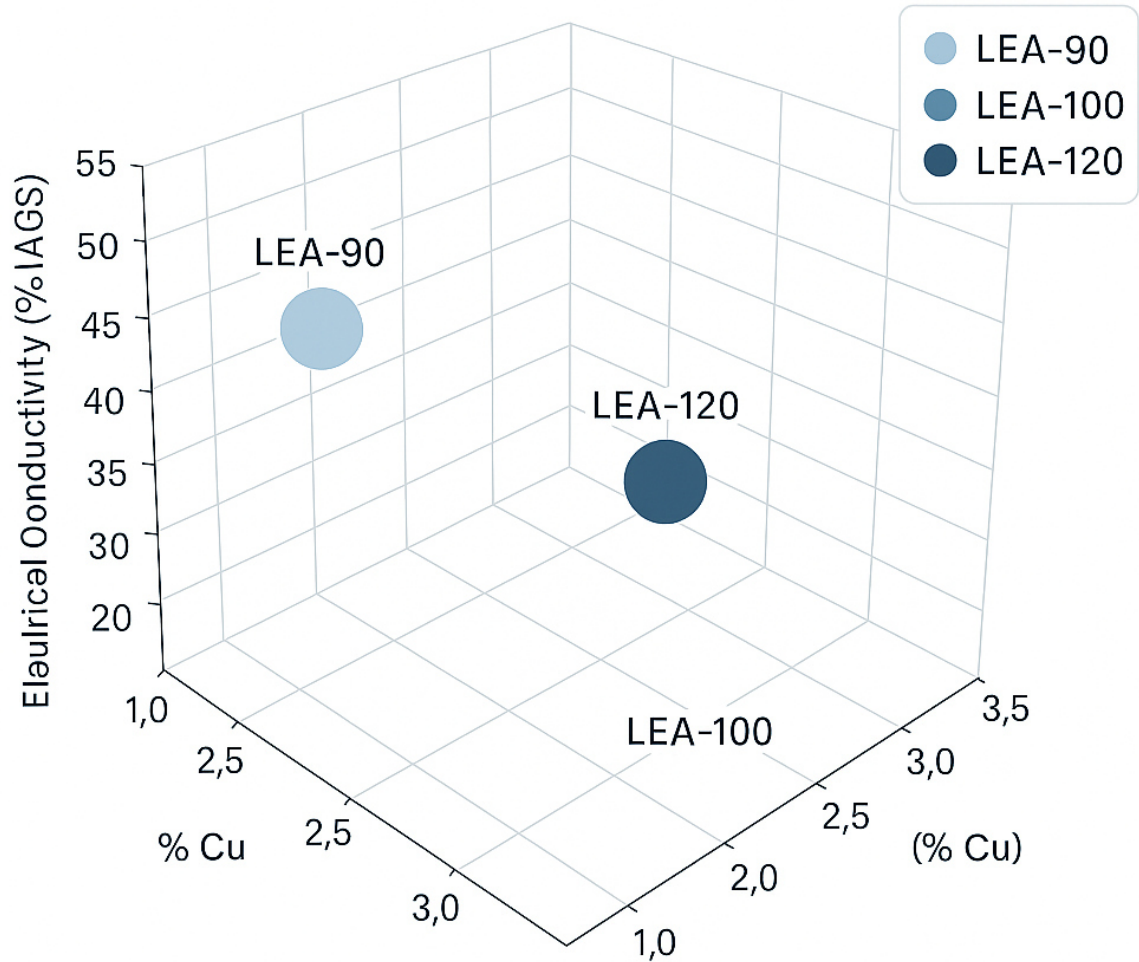
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LEA Alloy Series:

Electrical Conductivity vs. Tensile Strength vs. Cu Content



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